XAVIER BELTRAN URBANO

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PROFESSIONAL SUMMARY

PhD researcher in Bioengineering at the University of Pennsylvania with expertise in medical imaging, computer vision, and deep learning for neuroscience applications. Experienced in developing approaches for image segmentation, registration, and classification using Python and PyTorch. Holds a joint Master's degree in Medical Imaging and Artificial Intelligence from top European universities in France, Italy, and Spain. Passionate about advancing research in neurodegenerative diseases through intelligent algorithm design, real-time processing, and data-driven neuroimaging solutions.

EDUCATION

PhD Student in Bioengineering | GPA 4.0/4.0 **Ongoing**

University of Pennsylvania | Philadelphia, USA

Relevant Courses: Biomedical Image Analysis | Natural Language Processing | Applied Machine Learning | MRI | Alzheimer's and Dementia

Joint MSc in Medical Imaging and Artificial Intelligence | Grade 9.16/10 **Graduated**

University of Girona (Spain), University of Burgundy (France), University of Cassino (Italy)

Relevant Courses: Machine & Deep Learning | Image & Digital Signal Processing | Medical Image Segmentation, Registration & Classification

BSc in Biomedical Engineering | Grade 7.65/10 **Graduated**

University of Girona | Girona, Spain

Relevant Courses: Medical Image Analysis for Diagnosis | Neuroscience & Neuroimaging | Computer Vision | Artificial Intelligence

RESEARCH & WORK EXPERIENCE

Graduate Research Assistant (PhD) **August 2024 - Present**

Detrelab, University of Pennsylvania | Philadelphia, USA

- Spearheading the design of MARIA, a novel vision model leveraging multimodal MRI inputs and clinical data to predict the risk of developing Amyloid-Related Imaging Abnormalities (ARIA) before starting anti-amyloid treatment.
- Developed ICHOR, a self-supervised representation learning framework for ASL CBF maps using 3D masked autoencoders and Vision Transformers, pretrained on 11,405 multi-site scans to enable robust, generalizable feature extraction for downstream diagnostic and quality assessment tasks.
- Developed ART-Net, the first deep learning model that predicts the dominant artifact source in ASL CBF maps, enabling real-time artifact identification during acquisition and improving data quality and reproducibility.
- Leveraged advanced deep learning and computer vision techniques to generate explainable AI through 3D spatial heatmaps, enhancing model transparency and interpretability for clinical decision support.
- Developed a 3D GAN-based image-to-image translation framework for FDG-PET synthesis from multimodal MRI inputs to improve Alzheimer's disease diagnosis and monitoring (SSIM ≈ 0.94).

Medical Imaging / Machine Learning Intern **February 2024 – June 2024**

Detrelab, University of Pennsylvania | Philadelphia, USA

- Developed a Python-based user interface to streamline image quality rating workflows, enhancing efficiency and inter-rater consistency.
- Designed and implemented QEI-Net, the first deep learning model that assesses the quality of ASL CBF maps, achieving excellent performance and strong correlation with expert ratings (MSE = 0.009 | Pearson Correlation = 0.92).

Computer Vision Intern **July 2023 – October 2023**

R&D Department of icometrix**,** Leuven | Belgium

- Analyzed and harmonized a multicenter stroke brain imaging dataset containing CT perfusion maps.

- Developed a 3D deep learning-based post-processing approach (Accuracy = 93%) to remove artifacts in stroke CT perfusion maps.

Undergraduate Research Assistant **January 2022 – June 2022**

Computer Vision and Robotics Research Group (ViCOROB), University of Girona | Spain

- Applied both supervised and unsupervised machine learning algorithms to perform brain tumor segmentation (DSC ≈ 83%) from MRI images.
- Designed, developed, and 3D-printed a patient-specific skull and tumor model to support preoperative planning for brain surgery.

Undergraduate Research Assistant **June 2021 – September 2021**

Computer Vision and Robotics Research Group (ViCOROB), University of Girona | Spain

- Contributed to the design and development of various deep learning projects focused on dermatological imaging.
- Applied advanced computer vision techniques for melanoma detection in dermatological imaging, improving diagnostic accuracy and early detection.

SELECTED PUBLICATIONS AND CONFERENCES

- **X.Beltran-Urbano,** K.R. Jobson, S. Li, M. Taso, I.M. Nasrallah, P.A. Yushkevich, L. Xie, D.A. Wolk, S. Dolui, J.A. Detre, "SynthPET: A 3D Generative AI Approach For FDG-PET Image Synthesis from T1-Weighted MRI and ASL CBF Maps for Alzheimer's disease" (**Manuscript being prepared)**
- **X.Beltran-Urbano**, Y. Li, X. Zeng, K.R. Jobson, M. Taso, C.A. Brown, D.A. Wolk, C.T. McMillan, I.M. Nashrallah, P.A. Yushkevich, Z. Wang, J.A. Detre, S. Dolui, "ICHOR: A Robust Representation Learning Approach for ASL CBF Maps with Self-Supervised Masked Autoencoders" (**Submitted to MICCAI 2026 Conference in Abu Dhabi, UAE)** [link]
- **X.Beltran-Urbano,** M. Taso, K.R. Jobson, P.A. Yushkevich, I.M. Nasrallah, J.A. Detre, Z. Wang, S. Dolui, "ART-Net: A deep learning framework for artifact source classification in ASL CBF Maps" **(Accepted as digital poster in the ISMRM 2026 Conference in Cape Town, South Africa)** [link]
- **X.Beltran-Urbano**, M. Taso, I.M. Nasrallah, Z. Wang, J.A. Detre, S. Dolui, "QEI-Net: A deep learning-based automated quality evaluation index for ASL CBF Maps" (**Accepted as power-pitch presentation in the ISMRM 2025 Conference in Honolulu, Hawaii**) [link]
- **X.Beltran-Urbano**, M. Taso, I.M. Nasrallah, Z. Wang, J.A. Detre, S. Dolui, "QEI-Net: A Deep learning-based automated quality evaluation index for ASL CBF Maps" (**Accepted as an oral presentation in the ISMRM Workshop on Perfusion MRI in Pamplona, Spain**) [link]

SELECTED PROJECTS

A Hybrid Approach for Brain Tissue Segmentation: Integrating Gaussian Mixture Models with Atlas-based and Tissue Modeling Techniques

- Developed a hybrid brain MRI segmentation framework that combines Gaussian mixture models with atlas-based priors and tissue modeling to improve robustness and anatomical consistency.

Enhancing Brain Tissue Analysis: A 2D and 3D CNN Ensemble Approach for MRI Segmentation

- Designed an ensemble of 2D and 3D CNNs (U-Net, Res-U-Net, Dense-U-Net, and Seg-Res-U-Net) to leverage complementary spatial features for accurate and reliable brain tissue segmentation in MRI.

Skin Lesion Classification Using Deep Learning Techniques

- Developed a comprehensive skin lesion classification framework using the ISIC 2020 dataset, integrating classical machine learning models (Random Forest, SVM, KNN, XGBoost) with advanced deep learning architectures, including CNN-based models (ResNet, EfficientNet, DenseNet) and vision transformers (Swin Tiny, Swin-S), to perform robust binary and multiclass classification.

Breast Mass Detection System Based on a Multi-Scale Approach

- Developed an advanced computer vision pipeline for mammographic mass detection on the INbreast dataset, leveraging multi-scale morphological processing, superpixel segmentation (SLICO), and feature-based classification to achieve high sensitivity in identifying and delineating breast masses.

AWARDS AND RECOGNITIONS

- **The Peer Choice Award** by the Penn Nursing Smarter Care Datathon (*selected by participants as Best Project*) **2025**
- **The ISMRM Summa Cum Laude Merit Award** (*Top 5% of accepted abstracts*) **2025**
- **Twice awarded the prestigious INTHERAPI Graduate School Scholarship** by the University of Bourgogne **2022**, **2023**

- **Erasmus Mundus Joint Master Consortium Grant** by the University of Girona **2022**

RELEVANT SKILLS

Programming & Cloud Computing: Python | R | MATLAB | SQL | Java | Docker | Linux/Unix | Git/GitHub | LabVIEW | C++ | Arduino

Computer Vision & Deep Learning: PyTorch | TensorFlow | Keras | MONAI | Scikit-learn | OpenCV | SimpleITK

Imaging and Visualization Tools: 3D Slicer | ImageJ | ITK | Streamlit | Qt Designer | Photoshop

Core Expertise: 2D/3D Image Segmentation | Object Detection | Classification | Regression | Generative Models